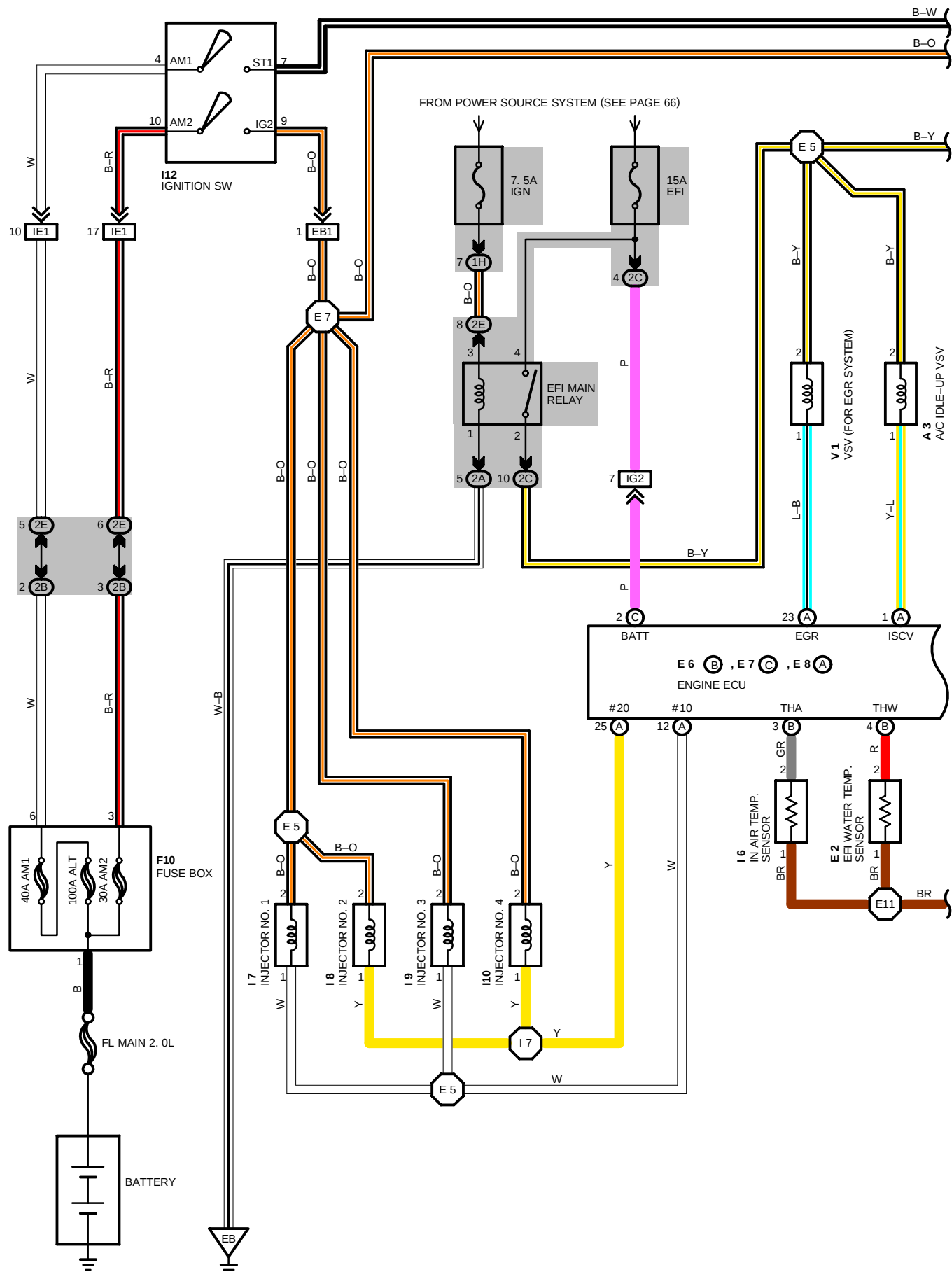
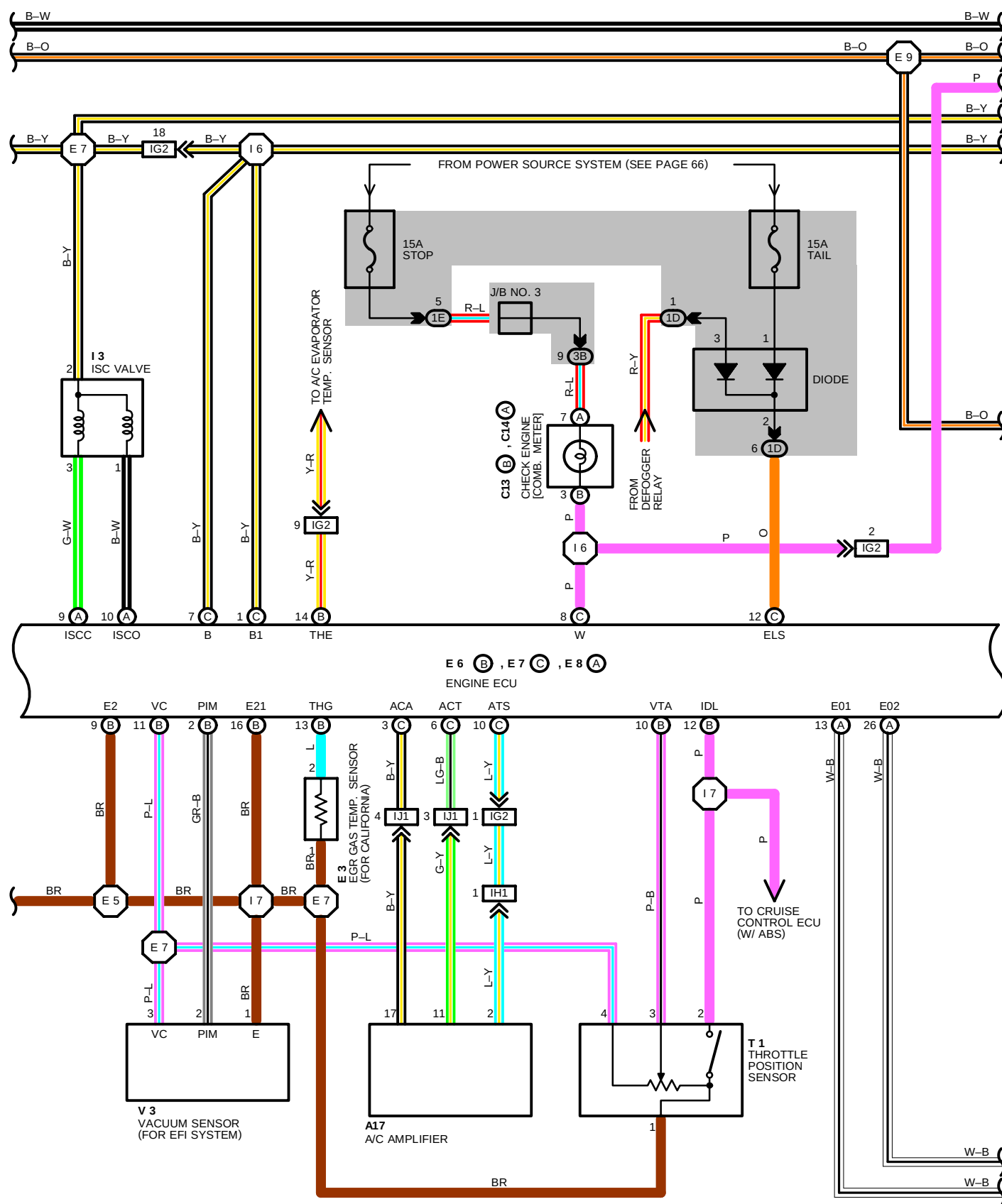
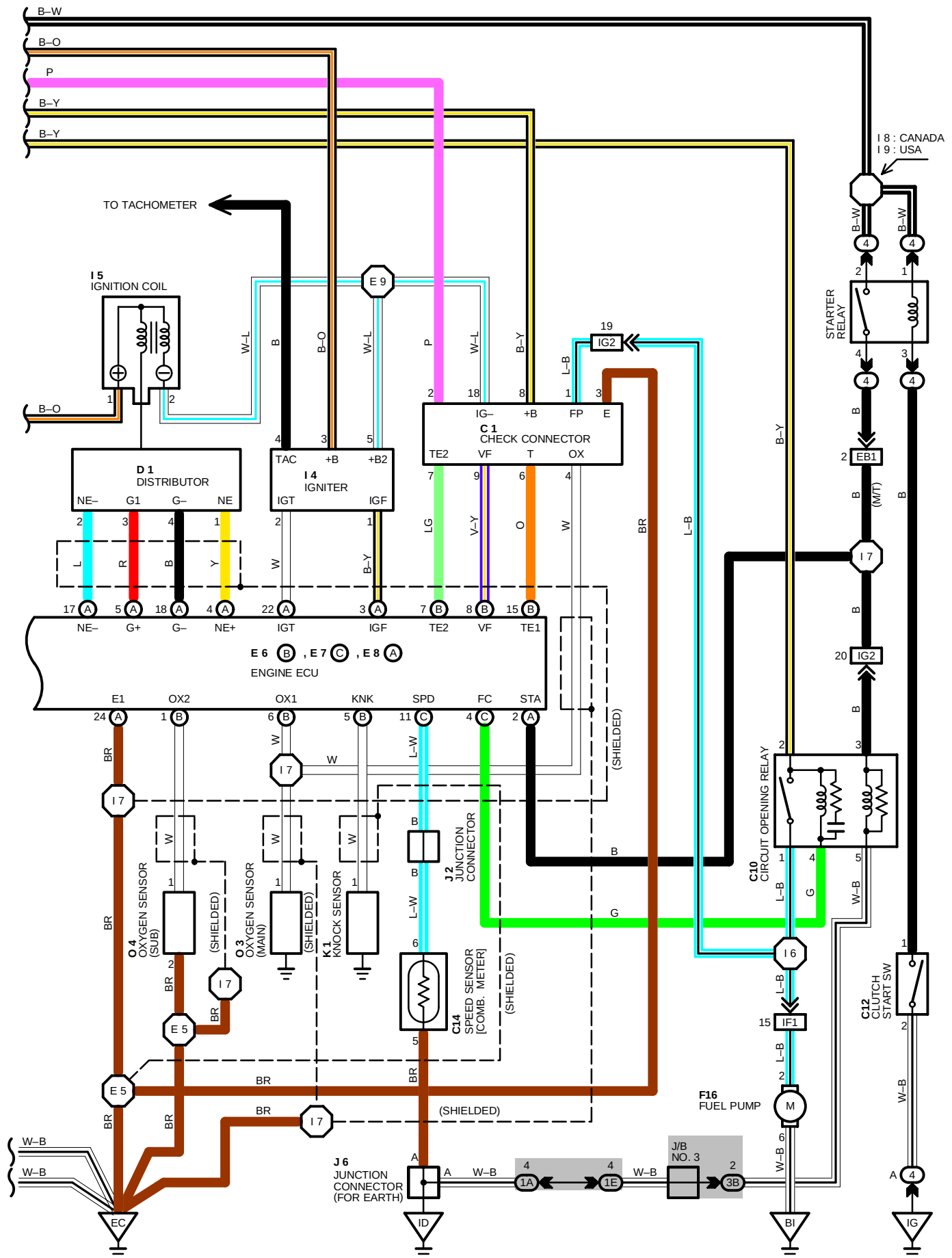


# ENGINE CONTROL (5S-FE M/T)





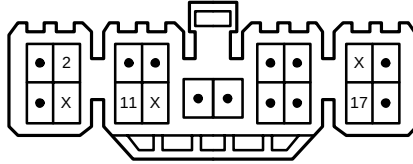
# ENGINE CONTROL (5S-FE M/T)



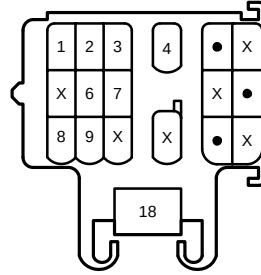
A 3 BLACK



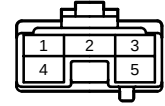
A17



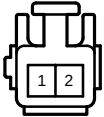
C 1 DARK GRAY



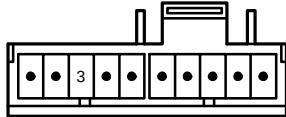
C10 DARK GREEN



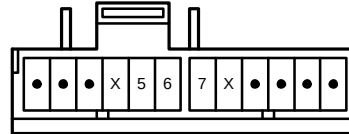
C12



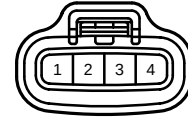
C13 (B) BROWN



C14 (A)



D 1 BLACK



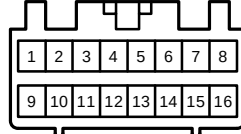
E 2 DARK GREEN



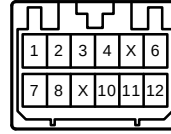
E 3 DARK GRAY



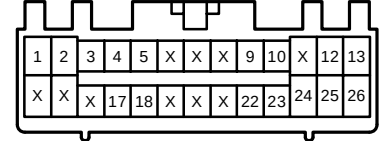
E 6 (B) DARK GRAY



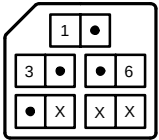
E 7 (C) DARK GRAY



E 8 (A) DARK GRAY



F10 BLACK



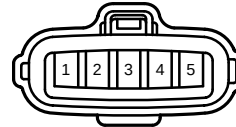
F16 DARK GRAY



I 3 GRAY



I 4 BLACK



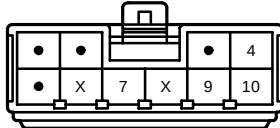
I 5 BLACK



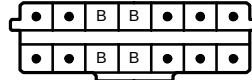
I 6 BLACK

I 7, I 8  
I 9, I 10 GRAY

I12 BLACK

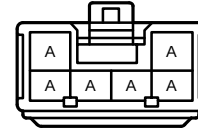


J 2



(HINT : SEE PAGE 7)

J 6

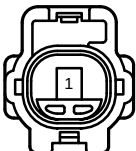


(HINT : SEE PAGE 7)

K 1 DARK GRAY



O 3 DARK GRAY



O 4 DARK GRAY



# ENGINE CONTROL (5S-FE M/T)

## SERVICE HINTS

### E 6, E 7, E 8 ENGINE ECU

VOLTAGE AT ECU WIRING CONNECTOR

|                     |                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|
| BATT-E1             | : ALWAYS 10.0-14.0 VOLTS                                                                       |
| +B-E1               | : 10.0-14.0 VOLTS (IGNITION SW AT <b>ON</b> POSITION)                                          |
| +B1-E1              | : 10.0-14.0 VOLTS (IGNITION SW AT <b>ON</b> POSITION)                                          |
| IDL-E1              | : 8.0-14.0 VOLTS (IGNITION SW ON AND THROTTLE VALVE OPEN)                                      |
| PSW-E1              | : 4.5-5.5 VOLTS (IGNITION SW ON AND THROTTLE VALVE FULLY CLOSED)                               |
| PIM-E2              | : 3.3-3.9 VOLTS (IGNITION SW AT <b>ON</b> POSITION)                                            |
| VC-E2               | : 4.5-5.5 VOLTS (IGNITION SW AT <b>ON</b> POSITION)                                            |
| #10, #20 - E01, E02 | : 10.0-14.0 VOLTS (IGNITION SW AT <b>ON</b> POSITION)                                          |
| THA-E2              | : 1.9-2.9 VOLTS (IGNITION SW ON AND INTAKE AIR TEMP. 20°C, 68°F)                               |
| THW-E2              | : 0.1-1.1 VOLTS (IGNITION SW ON AND COOLANT TEMP. 80°C, 176°F)                                 |
| STA-E1              | : 6.0-14.0 VOLTS (ENGINE CRANKING)                                                             |
| IGT-E1              | : 0.8-1.2 VOLTS (ENGINE CRANKING OR IDLING)                                                    |
| ISCC, ISCO-E1       | : 8.0-14.0 VOLTS (IGNITION SW AT <b>ON</b> POSITION)                                           |
| W-E1                | : 10.0-14.0 VOLTS (IGNITION SW ON, NO TROUBLE AND ENGINE RUNNING)                              |
| ACT-E1              | : 4.5-5.5 VOLTS (IGNITION SW ON AND AIR CONDITIONING ON)                                       |
| ACA-E1              | : 2.0 VOLTS OR LESS (IGNITION SW ON AND AIR CONDITIONING ON)                                   |
| TE1-E1              | : 10.0-14.0 VOLTS (IGNITION SW ON AND CHECK CONNECTOR T-E1 NOT CONNECTED)                      |
|                     | : 1.0 VOLTS OR LESS (IGNITION SW ON AND CHECK CONNECTOR T-E1 CONNECTOR)                        |
| NSW-E1              | : 0-2.0 VOLTS (IGNITION SW ON AND NEUTRAL START SW POSITION <b>P</b> OR <b>N</b> RANGE)        |
|                     | : 6.0-14.0 VOLTS (IGNITION SW ON AND EX. NEUTRAL START SW POSITION <b>P</b> OR <b>N</b> RANGE) |

### RESISTANCE AT ECU WIRING CONNECTORS

(DISCONNECT WIRING CONNECTOR)

|                    |                                            |
|--------------------|--------------------------------------------|
| IDL-E1             | : INFINITY (THROTTLE VALVE OPEN)           |
|                    | : 0 Ω (THROTTLE VALVE FULLY CLOSED)        |
| PSW-E1             | : 0 Ω (THROTTLE VALVE FULLY OPEN)          |
|                    | : INFINITY (THROTTLE VALVE FULLY CLOSED)   |
| THA-E2             | : 2.0-3.0 KΩ (INTAKE AIR TEMP. 20°C, 68°F) |
| THW-E2             | : 0.2-0.4 KΩ (COOLANT TEMP. 80°C, 176°F)   |
| G+ - G-            | : 0.17-0.21 KΩ                             |
| ISCC, ISCO-+B, +B1 | : 19.3-22.3 KΩ                             |

## : PARTS LOCATION

| CODE | SEE PAGE   | CODE | SEE PAGE           | CODE | SEE PAGE   |
|------|------------|------|--------------------|------|------------|
| A 3  | 30         | E 7  | C 33               | I10  | 30 (5S-FE) |
| A17  | 32         | E 8  | A 33               | I12  | 33         |
| C 1  | 30 (5S-FE) | F10  | 30 (5S-FE)         | J 2  | 33         |
| C10  | 32         | F16  | 34 (L/B), 35 (C/P) | J 6  | 33         |
| C12  | 32         | I 3  | 30 (5S-FE)         | K 1  | 30         |
| C13  | B 32       | I 4  | 30 (5S-FE)         | O 3  | 30 (5S-FE) |
| C14  | A 32       | I 5  | 30 (5S-FE)         | O 4  | 30 (5S-FE) |
| D 1  | 30 (5S-FE) | I 6  | 30 (5S-FE)         | T 1  | 30 (5S-FE) |
| E 2  | 30 (5S-FE) | I 7  | 30 (5S-FE)         | V 1  | 30 (5S-FE) |
| E 3  | 30 (5S-FE) | I 8  | 30 (5S-FE)         | V 3  | 30 (5S-FE) |
| E 6  | B 33       | I 9  | 30 (5S-FE)         |      |            |

## : RELAY BLOCKS

| CODE | SEE PAGE | RELAY BLOCKS (RELAY BLOCK LOCATION) |
|------|----------|-------------------------------------|
| 4    | 27       | R/B NO. 4 (RIGHT KICK PANEL)        |

## : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

| CODE | SEE PAGE | JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)   |
|------|----------|--------------------------------------------------------|
| 1A   | 20       | COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)              |
| 1D   |          |                                                        |
| 1E   |          |                                                        |
| 1H   | 20       | ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)  |
| 2A   | 22 (2WD) | ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY) |
| 2B   | 22 (2WD) | ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)           |
| 2C   |          |                                                        |
| 2E   | 22 (2WD) | ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY) |
| 3B   | 25       | COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)     |

 : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

| CODE | SEE PAGE   | JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)  |
|------|------------|-------------------------------------------------------------|
| EB1  | 40 (5S-FE) | ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER) |
| IE1  | 44         | ENGINE ROOM MAIN WIRE AND COWL WIRE (LEFT KICK PANEL)       |
| IF1  | 44         | FLOOR WIRE AND COWL WIRE (LEFT KICK PANEL)                  |
| IG2  | 44         | ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE ECU)            |
| IH1  | 44         | ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND GLOVE BOX WIRE)      |
| IJ1  | 46         | COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)         |

 : GROUND POINTS

| CODE | SEE PAGE   | GROUND POINTS LOCATION       |
|------|------------|------------------------------|
| EB   | 40 (5S-FE) | FRONT LEFT FENDER            |
| EC   | 40 (5S-FE) | INTAKE MANIFOLD              |
| ID   | 44         | LEFT KICK PANEL              |
| IG   | 44         | R/B NO. 4 SET BOLT           |
| BI   | 48 (L/B)   | UNDER THE LEFT CENTER PILLAR |
|      | 50 (C/P)   |                              |

 : SPLICE POINTS

| CODE | SEE PAGE   | WIRE HARNESS WITH SPLICE POINTS | CODE | SEE PAGE | WIRE HARNESS WITH SPLICE POINTS |
|------|------------|---------------------------------|------|----------|---------------------------------|
| E 5  | 40 (5S-FE) | ENGINE WIRE                     | I 6  | 46       | COWL WIRE                       |
| E 7  |            |                                 | I 7  | 46       | ENGINE WIRE                     |
| E 9  |            |                                 | I 8  | 46       | COWL WIRE                       |
| E11  |            |                                 | I 9  | 46       |                                 |